

Exercise 1 – HTTP Requests 0,1 Points

- a. Use the *Developer Tools* of your web browser (e.g., Chrome) to inspect the HTTP requests when opening the website of www.aau.at and answer the following questions:
- How many requests are sent, how much data is transferred, and how long does it take to load the site from the server?
173 requests, 7,7 MB transferred and 3,22s Load and after 3,68s Finished
 - What is the *response status code* of the first request?
200
 - What type of web server is serving this website?
nginx
 - What types of resources are requested?
JPEG, PNG, Document, Script, Font, Stylesheet
 - Which request is the slowest?
The Blog-Beitrag-Foto-845x684.png GET Request is the slowest
 - Which cookies does the site store?
aviaCookieConsent, aviaPrivacyEssentialCookiesEnabled, aviaPrivacyMustOptInSetting, aviaPrivacyRefuseCookiesHideBar
- b. Repeat the procedure and answer the same questions for www.orf.at
- How many requests are sent, how much data is transferred, and how long does it take to load the site from the server?
86 requests, 2,3 MB transferred and 2,33s Load and Finished 28,43s
 - What is the *response status code* of the first request?
204
 - What type of web server is serving this website?
Apache
 - What types of resources are requested?
JPEG, PNG, Document, Script, Font, Stylesheet, Fetch, Ping, XHR, GIF, svg+xml, webp, Preflight
 - Which request is the slowest?
The ORFUniversalVF_W_WghtWdthOpsz.woff2 GET Request is the slowest
 - Which cookies does the site store?
_pk_id.5.186e, _pk_ses.5.186e, didomi_token, euconsent-v2, i00, ioam2018
- c. Open up www.google.at, clear the content of the network inspection window, and enter the word “Klagenfurt” into the search field. While entering the word, look at the network requests and explain what happens.
After every letter the Url gets updated:
https://www.google.com/complete/s?q=k&cp=1&client=gws-wiz&xssi=t&gs_pcr=undefined&hl=de-AT&authuser=0&psi=Gn3HaubGAsqJ7NYPiLWDsAE.1791458587630&dpr=1.25

d. What is the cache?

A cache is temporary storage used by a browser to save copies of resources that have already been downloaded.

How can you delete it?

In Chrome: STRG+Shift+Entf -> Then click “Von diesem Gerät löschen“

When could it be useful to disable or delete the cache?

Website has been updated, isn't displaying correctly, When developing/testing a site, free some storage space

Exercise 2 – Setting up Apache 0,1 Points

Install Apache HTTP Server version 2.4 and open <http://localhost> in your browser. You should see the text “It works”.

Additionally, answer the following questions:

- Describe the purpose of `conf/httpd.conf` and explain the meaning of the following terms: `ServerRoot`, `DocumentRoot`, `Virtual Host`

`httpd.conf`: Main Apache configuration file

`ServerRoot`: Main directory of the Apache installation

`DocumentRoot`: Directory containing files served by the website

`Virtual Host`: Allows one Apache server to host multiple websites

- What is the loopback IP address 127.0.0.1 and the hostname localhost used for?

127.0.0.1: IPv4 loopback address referring to the local computer

Localhost: Hostname normally referring to the local computer/loopback interface

- What is the purpose of the `index.html` document?

It contains the content of the <http://localhost> website.

Find the document, edit its content, and open <http://localhost> again. Create a folder “mytest” in the documents root folder and add some random files and folders to the new folder. Open <http://localhost/mytest> in your browser and explain the result.

The result is that you can see a list of the files and folders. You can also download the files from the folder.

Exercise 4

Make sure to use the `alt` attribute in a meaningful way. Why is this attribute and a meaningful value for it important?

If there is a problem with the image, the alternative text can be displayed instead.

Exercise 7 - CSS Selectors 0,1 Points

Consider the HTML Structure given in file *ex7.html*. Explain what happens if you apply separately each of the following style definitions to the HTML structure. Uncomment one rule at a time in file *ex7.html* and explain the result:

Note: You can use the developer console of your web browser to apply the given CSS rules.

`#content, p { color: red; }` = Die ID mit dem Namen „content“ und alle `<p>` Elemente sind rot

`#content p { color: red; }` = Alle `<p>` Elemente in der ID content sind, sind rot

`#content h2 p { color: red; }` = Alle `<p>` Elemente in `<h2>` Elementen, welche auch in ID content sind, sind rot

`#content > p { color: red; }` = Alle `<p>` Elemente direct unter der ID content sind rot

`#content h1 p { color: red; }` = Alle `<p>` Elemente in `<h1>` Elementen, welche auch in ID content sind, sind rot

`div p:nth-child(2) { color: red; }` = Alle zweiten Kinderelemente vom `<p>` Element innerhalb der `<div>` Elemente sind rot, auch diejenigen, die tiefer in den Elementen verschachtelt sind

`div p:nth-child(2n) { color: red; }` = Alle geraden Kinderelemente in allen Verschachtelungsebenen innerhalb der `<div>` Elemente sind rot

`h2 {color: red;}` = jedes `<h2>` Element ist rot

`div.othercontent > p:first-child { color:red; }` = Wenn das erste Kinderelement direkt unterhalb von `div.othercontent` ein `<p>` Element ist, ist es rot.

`div.othercontent h1 + p { color:red; }` = Das erste `<p>` Element in `div.othercontent` nach einem `h1`-Element ist rot

`div.othercontent h1 ~ p { color:red; }` = Alle `<p>` Elemente in `div.othercontent` nach einem `<h1>` Element sind rot

`a[title^=other]:link { color:red; }` = Alle nicht besuchten `<a>` Elemente, deren title-Attribut mit „other“ beginnt, sind rot

`body a[title^=other]:link { color:red; }` = Alle nicht besuchten `<a>` Elemente in `<body>`, deren title-Attribut mit „other“ beginnt, sind rot

`.othercontent * p { color: red; }` = Alle `<p>` Elemente in der Klasse `<othercontent>`, die nicht direkt unter `<othercontent>` sind, sind rot.